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Laboratório 5: Laboratório Especial

Para desenvolver o código do mapa, tive o auxílio de ferramentas de IA, principalmente com o objetivo de entender a sintaxe e criar meu próprio código.

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(readr)
library(leaflet)
analisa_aeronave <- function(tail_number, arquivo) {
  funcao_callback <- function(input, pos) {
    input %>% filter(TAIL_NUMBER == tail_number)
  }
  tabela_base <- read_csv_chunked(file = arquivo,
                                callback = DataFrameCallback$new(funcao_callback),
                                chunk_size = 1e5)
  base_aeroportos <- read_csv("airports.csv") %>%
    select(IATA_CODE, LATITUDE, LONGITUDE)
  tabela_tratada <- tabela_base %>%
    arrange(YEAR, MONTH, DAY, SCHEDULED_DEPARTURE) %>%
    left_join(base_aeroportos, by = c("ORIGIN_AIRPORT" = "IATA_CODE")) %>%
    rename(lat_orig = LATITUDE, lon_orig = LONGITUDE) %>%
    left_join(base_aeroportos, by = c("DESTINATION_AIRPORT" = "IATA_CODE")) %>%
    rename(lat_dest = LATITUDE, lon_dest = LONGITUDE) %>%
    filter(!is.na(AIR_TIME), !is.na(lat_orig), !is.na(lon_orig), !is.na(lat_dest), !is.na(lon_dest))
    mutate(velocidade_media = DISTANCE / (AIR_TIME / 60))
  mapa_trajeto <- leaflet() %>% addTiles()
  for(i in 1:nrow(tabela_tratada)) {
    voo_atual <- tabela_tratada[i, ]
    espessura_linha <- voo_atual$velocidade_media / 80
    mapa_trajeto <- mapa_trajeto %>%
      addPolylines(
        lng = c(voo_atual$lon_orig, voo_atual$lon_dest),
        lat = c(voo_atual$lat_orig, voo_atual$lat_dest),
        weight = espessura_linha,
        color = "#1f78b4",
```

```

    opacity = 0.7
  ) %>%
  addCircleMarkers(
    lng = voo_atual$lon_orig,
    lat = voo_atual$lat_orig,
    radius = 1,
    color = "red"
  )
}
return(list(
  tabela = tabela_tratada,
  grafico = mapa_trajeto
))
}
resultado_final <- analisa_aeronave(tail_number = "N431WN", arquivo = "flights.csv")

```

— Column specification —

```

cols(
  .default = col_double(),
  AIRLINE = col_character(),
  TAIL_NUMBER = col_character(),
  ORIGIN_AIRPORT = col_character(),
  DESTINATION_AIRPORT = col_character(),
  SCHEDULED_DEPARTURE = col_character(),
  DEPARTURE_TIME = col_character(),
  WHEELS_OFF = col_character(),
  WHEELS_ON = col_character(),
  SCHEDULED_ARRIVAL = col_character(),
  ARRIVAL_TIME = col_character(),
  CANCELLATION_REASON = col_character()
)

```

• Use `spec()` for the full column specifications.

Rows: 322 Columns: 7

— Column specification —

Delimiter: ","

chr (5): IATA_CODE, AIRPORT, CITY, STATE, COUNTRY

dbl (2): LATITUDE, LONGITUDE

• Use `spec()` to retrieve the full column specification for this data.

• Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
resultado_final$tabela
```

A tibble: 1,764 × 36

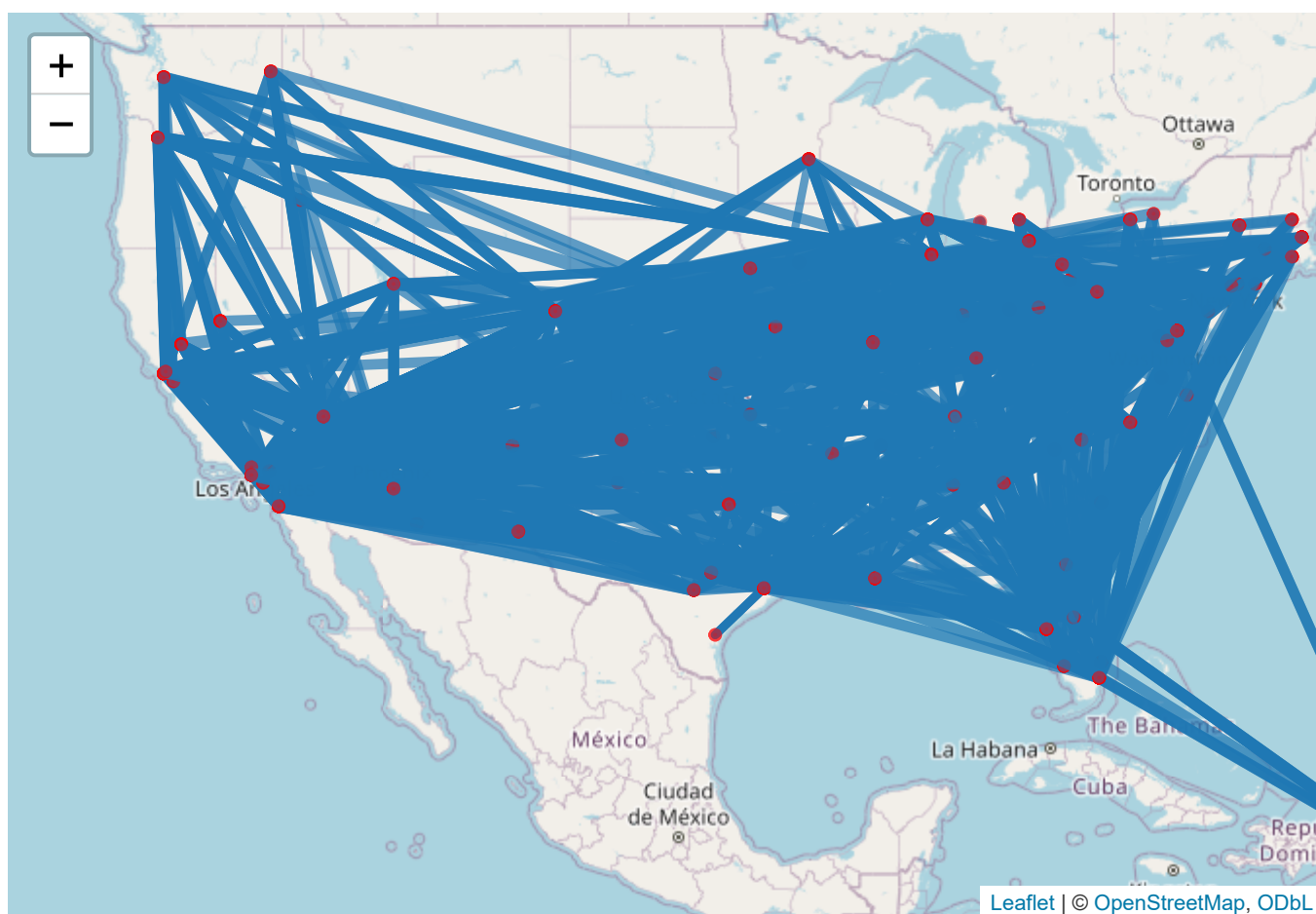
	YEAR	MONTH	DAY	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER
	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>	<chr>
1	2015	1	1	4	WN	1762	N431WN
2	2015	1	1	4	WN	3564	N431WN
3	2015	1	1	4	WN	3564	N431WN
4	2015	1	1	4	WN	3564	N431WN

```

5  2015    1    2        5 WN        1793 N431WN
6  2015    1    2        5 WN        1793 N431WN
7  2015    1    2        5 WN        1793 N431WN
8  2015    1    2        5 WN        1793 N431WN
9  2015    1    2        5 WN        1793 N431WN
10 2015    1    2        5 WN        3144 N431WN
# i 1,754 more rows
# i 29 more variables: ORIGIN_AIRPORT <chr>, DESTINATION_AIRPORT <chr>,
#   SCHEDULED_DEPARTURE <chr>, DEPARTURE_TIME <chr>, DEPARTURE_DELAY <dbl>,
#   TAXI_OUT <dbl>, WHEELS_OFF <chr>, SCHEDULED_TIME <dbl>, ELAPSED_TIME <dbl>,
#   AIR_TIME <dbl>, DISTANCE <dbl>, WHEELS_ON <chr>, TAXI_IN <dbl>,
#   SCHEDULED_ARRIVAL <chr>, ARRIVAL_TIME <chr>, ARRIVAL_DELAY <dbl>,
#   DIVERTED <dbl>, CANCELLED <dbl>, CANCELLATION_REASON <chr>, ...

```

```
resultado_final$grafico
```



Horário de compilação do arquivo:

```

Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()

```

```
[1] "2026-09-16 22:08:21 -03"
```